

Sagnik Nandy

Curriculum Vitae

Department of Statistics
The Ohio State University
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Work Experience

- 2025– **Assistant Professor**, *The Ohio State University*, Columbus, USA.
2024–2025 **Kruskal Instructor**, *University of Chicago*, Chicago, USA.

Education

- 2019–2024 **PhD in Statistics**, *University of Pennsylvania*, Philadelphia, USA.
◦ Advised by Zongming Ma and Bhaswar B. Bhattacharya.
◦ Dissertation: Orchestrated Approximate Message Passing: A Novel Way of Multimodal Data Integration.
2017–2019 **Master of Statistics**, *Indian Statistical Institute*, Kolkata, India.
◦ Specialization: Theoretical Statistics.
2014–2017 **Bachelor of Statistics (Hons.)**, *Indian Statistical Institute*, Kolkata, India.

Awards and Honors

- 2023 J. Parker Memorial Bursk Prize for Excellence in Research, [Department of Statistics, University of Pennsylvania](#),
Awarded for excellence in research during the PhD program.
2019 PC Mahalanobis Gold Medal, [Indian Statistical Institute](#),
Awarded for exceptional academic performance during the program.
2019 J.K. Ghosh Gold Medal, [Indian Statistical Institute](#),
Awarded to the topper of the program.
2017 D. Basu Gold Medal, [Indian Statistical Institute](#),
Awarded for exceptional academic performance during the program.
2017 Ms. M. S. Iyer Memorial Gold Medal, [Indian Statistical Institute](#),
Awarded to the topper of the program.
2017 Nikhilesh Bhattacharya Gold Medal, [Indian Statistical Institute](#),
Awarded to the students with highest grades in Statistics.
2014 INSPIRE scholarship, [Department of Science and Technology, Government of India](#),
Awarded to top 1% students in the Indian School Certificate Examinations, 2014.

Research Interest

High-Dimensional and Non-Parametric Statistics and their Applications in Single-cell Genomics, Inference in Graphical models, Random Matrix Theory and its Applications.

Publications

- 2023 **Community Detection With Contextual Multilayer Networks.**
Zongming Ma, Sagnik Nandy
IEEE Transactions on Information Theory 69.5 (2023) pp. 3203–3239. 2023
<https://ieeexplore.ieee.org/document/10023524>

- 2023 **Exact Detection Thresholds for Chatterjee's Correlation.**
 Arnab Auddy, Nabarun Deb, Sagnik Nandy
Bernoulli 30 (2), 1640-1668
<https://arxiv.org/abs/2104.15140>
- 2024 **Degree Heterogeneity in Higher-Order Networks: Inference in the Hypergraph β -Model.**
 Sagnik Nandy, Bhaswar B. Bhattacharya
IEEE Transactions on Information Theory 70 (8), 6000-6024
<https://ieeexplore.ieee.org/abstract/document/10552775>
- 2024 **Detecting planted partition in sparse multilayer networks.**
 Anirban Chatterjee, Sagnik Nandy, Ritwik Sadhu
Information and Inference: A Journal of the IMA 13 (3)
<https://academic.oup.com/imaiai/article-abstract/13/3/iaae019/7726402>
- 2025 **Bayes optimal learning in high-dimensional linear regression with network side information.**
 Sagnik Nandy, Subhabrata Sen
IEEE Transactions on Information Theory 71 (8), 565-591
<https://ieeexplore.ieee.org/abstract/document/10736623>
- 2025 **Extremal Eigenvalues of Random Kernel Matrices with Polynomial Scaling.**
 David Kogan, Sagnik Nandy, Jiaoyang Huang
Random Matrices: Theory and Applications 14 (4), 2550020
<https://www.worldscientific.com/doi/10.1142/S2010326325500200>
- 2026 **Clustering by Denoising: Latent plug-and-play diffusion for single-cell data.**
 Dominik Meier, Shixing Yu, Sagnik Nandy, Promit Ghosal, Kyra Gan
The Fourteenth International Conference on Learning Representations, 2026
<https://arxiv.org/abs/2510.22835>
- 2026 **How Private is Your Attention? Bridging Privacy with In-Context Learning.**
 Soham Bonnerjee, Zhen Wei, Anna Asch, Sagnik Nandy, Promit Ghosal
Transactions on Machine Learning Research, 2026
<https://openreview.net/forum?id=M2qsrIbaOL>
- 2024+ **PriME: Privacy-aware Membership profile Estimation in networks.**
 Abhinav Chakraborty, Sayak Chatterjee, Sagnik Nandy
Under revision in Siam Journal on Mathematics of Data Science
<https://arxiv.org/abs/2406.02794>
- 2024+ **Multimodal data integration and cross-modal querying via orchestrated approximate message passing.**
 Sagnik Nandy, Zongming Ma
<https://arxiv.org/abs/2407.19030>
- 2026+ **Privacy utility trade offs for parameter estimation in degree heterogeneous higher order networks.**
 Bibhabasu Mandal, Sagnik Nandy
Submitted to COLT 2026
<https://arxiv.org/abs/2602.03948>
- 2026+ **How does limma-trend work? An empirical partially Bayes perspective.**
 Sagnik Nandy, Wanyi Ling, Nikolaos Ignatiadis
Submitted to JASA
<https://arxiv.org/abs/2605.21304>
- 2026+ **Transfer Learning in High-Dimensional Clustering: Minimax Thresholds and Applications in Single-Cell Data.**
 Abhinav Chakraborty, Sagnik Nandy
<https://arxiv.org/abs/2607.25031>
- 2026+ **DAIF: A Data-Driven Intermediate Fusion Framework for Multimodal Supervised Learning via Approximate Message Passing.**
 Sagnik Nandy, Samriddha Lahiry, Pragya Sur, Subhabrata Sen
<https://arxiv.org/abs/2608.02769>

Talks, Conferences and Workshops

- August, 2026 **Invited Talk**, *27th International Conference on Computational Statistics*, Athens, Greece.
Title: Privacy-aware Membership profile Estimation in networks
- July, 2026 **Invited Talk**, *Department of Mathematics, Indian Institute of Technology*, Mumbai, India.
Title: How does limma-trend work? An empirical partially Bayes perspective
- December, 2025 **Invited Talk**, *CFE-CM Statistics Conference*, London, UK.
Title: Degree Heterogeneity in Higher-Order Networks: Inference in Hypergraph β -model
- August, 2025 **Invited Talk**, *Department of Mathematics, Indian Institute of Technology*, Kanpur, India.
Title: Orchestrated Approximate Message Passing: A New Way Of Information Integration From Multimodal Data
- July, 2025 **Invited Talk**, *Department of Mathematics, Indian Institute of Technology*, Mumbai, India.
Title: Orchestrated Approximate Message Passing: A New Way Of Information Integration From Multimodal Data
- June, 2025 **Invited Talk**, *INFORMS Applied Probability Conference*, Atlanta, Georgia.
Title: Degree Heterogeneity in Higher-Order Networks: Inference in Hypergraph β -model
- June, 2025 **Invited Talk**, *IISA Annual Conference*, Lincoln, Nebraska.
Title: Extremal Eigenvalues of Random Kernel Matrices with Polynomial Scaling
- December, 2024 **Invited Talk**, *IISA Annual Conference*, Kochi, India.
Title: Community Detection With Contextual Multilayer Networks
- December, 2024 **Invited Talk**, *CFE-CM Statistics Conference*, London, UK.
Title: Detecting planted partition in sparse multilayer networks
- August, 2024 **Contributed Talk**, *IMS-Bernoulli World Congress in Statistics and Probability*, Bochum, Germany.
Title: Bayes optimal learning in high-dimensional linear regression with network side information
- March, 2024 **Invited Talk**, *Probabilitas Seminar, Harvard University*, Cambridge, Massachusetts.
Title: Orchestrated Approximate Message Passing: A New Way Of Information Integration From Multimodal Data
- January, 2024 **Invited (Job) Talk**, *Department of Statistics, University of Pittsburgh*, Pittsburgh, Pennsylvania.
Title: Orchestrated Approximate Message Passing: A New Way Of Information Integration From Multimodal Data
- January, 2024 **Invited (Job) Talk**, *Department of Statistics, The Ohio State University*, Columbus, Ohio.
Title: Orchestrated Approximate Message Passing: A New Way Of Information Integration From Multimodal Data
- January, 2024 **Invited (Job) Talk**, *Marshall School of Business, University of Southern California*, Los Angeles, California.
Title: Orchestrated Approximate Message Passing: A New Way Of Information Integration From Multimodal Data
- January, 2024 **Invited (Job) Talk**, *Department of Statistics, Florida State University*, Tallahassee, Florida.
Title: Orchestrated Approximate Message Passing: A New Way Of Information Integration From Multimodal Data
- December, 2023 **Contributed Talk**, *International Conference on Statistics and Data Science*, Lisbon, Portugal.
Title: Degree Heterogeneity in Higher-Order Networks: Inference in Hypergraph β -model
- August, 2023 **Invited Talk**, *Joint Statistical Meeting*, Toronto, Canada.
Title: Community Detection with Contextual Multilayer Networks
- July, 2023 **Invited Talk**, *Stat-Math Seminar, ISI Delhi*, Delhi, India.
Title: Orchestrated Approximate Message Passing: A New Way Of Information Integration From Multimodal Data
- July, 2023 **Invited Talk**, *Department of Mathematics, IIT Delhi*, Delhi, India.
Title: Orchestrated Approximate Message Passing: A New Way Of Information Integration From Multimodal Data
- January, 2023 **Invited Talk**, *Applied Statistics Unit Seminar, ISI Kolkata*, Kolkata, India.
Title: Orchestrated AMP in Network-Constrained Regression
- June, 2022 **Contributed Talk**, *IMS Annual Meeting*, London, UK.
Title: Community Detection with Contextual Multilayer Networks

September, **Invited Talk**, *Stat-Math Seminar, ISI Kolkata*, Kolkata, India.
2021 Title: Community Detection with Contextual Multilayer Networks

Teaching Experience

- Instructor for Stat 6201 (Graduate Course in Mathematical Statistics) at the Ohio State University for Fall 2026.
- Instructor for Stat 3302 (Undergraduate Course in Statistical Modelling for Discovery, Part II) at the Ohio State University for Fall 2025.
- Instructor for Stat 6201 (Graduate Course in Mathematical Statistics) at the Ohio State University for Fall 2025.
- Instructor for Stat 22400 (Undergraduate Course in Regression Techniques) at the University of Chicago for Spring 2025.
- Instructor for Stat 41500 (Graduate Course in High Dimensional Statistics) at the University of Chicago for Fall 2024.
- Teaching assistant for Stat 431 (Undergraduate Course in Mathematical Statistics) at the University of Pennsylvania for Spring 2023, Fall 2022, and Spring 2022.
- Teaching assistant for Stat 970 (Ph.D. Level course in Mathematical Statistics) at the University of Pennsylvania for Fall 2021.
- Teaching assistant for Stat 101 (Undergraduate Introductory Course in Statistics) at the University of Pennsylvania for Fall 2020 and Spring 2021.
- Graduate Student instructor in the Undergraduate Research in Statistics and Probability program for David Kogan with Dr. Jiaoyang Huang.

Computer skills

Programming Languages Python, C, R, MATLAB

Professional Service

- 2019 Organized Statistics Department Lunch Seminar
- 2023 Graduate student mentor for incoming first-year PhD students
- 2019-2025 Reviewer for Journal of American Statistical Association, Bernoulli, Journal of Machine Learning Research, Annals of Statistics, IEEE Transactions on Information Theory.
- 2025 Member of the dissertation committee of Qian Zhou and Xiangni Peng at OSU.
- 2025 Member of the graduate admissions committee at OSU.
- 2025 Member of the qualifying examinations committee at OSU.

References

- Bhaswar B. Bhattacharya, Associate Professor of Statistics and Data Science, University of Pennsylvania,
email: bhaswar@wharton.upenn.edu
- Zongming Ma, Professor of Statistics and Data Science, Yale University,
email: zongming.ma@yale.edu
- Subhabrata Sen, Assistant Professor of Statistics, Harvard University,
email: subhabratasen@fas.harvard.edu
- Jiaoyang Huang, Assistant Professor of Statistics and Data Science, University of Pennsylvania,
email: huangjy@wharton.upenn.edu
- Nikos Ignatiadis, Assistant Professor of Statistics, University of Chicago,
email: ignat@uchicago.edu